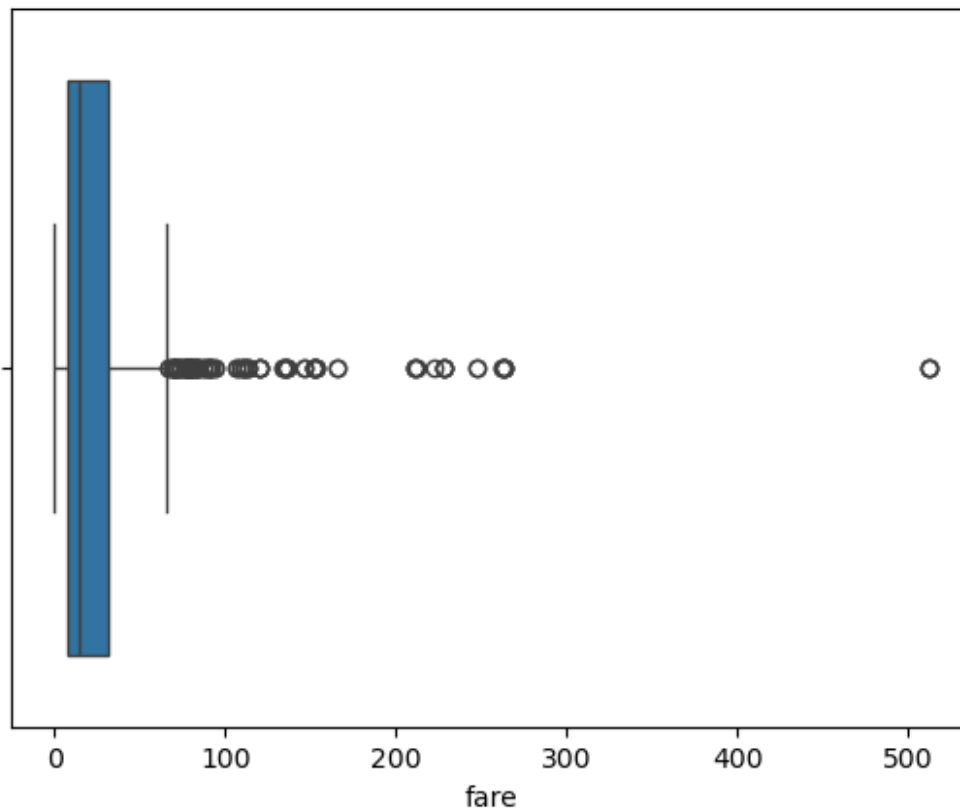


```

import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from scipy.stats import zscore
df = sns.load_dataset('titanic')
sns.boxplot(x=df['fare'])
plt.show()
Q1 = df['fare'].quantile(0.25)
Q3 = df['fare'].quantile(0.75)
IQR = Q3 - Q1
lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR
outliers_iqr = df[
    (df['fare'] < lower) |
    (df['fare'] > upper)]
print("IQR Outliers:", len(outliers_iqr))

#z-score method
df['fare_zscore'] = zscore(df['fare'])
outliers_z = df[
    abs(df['fare_zscore']) > 3
]
print("Z-score Outliers:", len(outliers_z))

```



IQR Outliers: 116
Z-score Outliers: 20

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
df = sns.load_dataset("tips")
print(df.head())
print(df.info())
print(df.columns)
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 244 entries, 0 to 243

Data columns (total 7 columns):

#	Column	Non-Null Count	Dtype
0	total_bill	244 non-null	float64
1	tip	244 non-null	float64
2	sex	244 non-null	category
3	smoker	244 non-null	category
4	day	244 non-null	category
5	time	244 non-null	category
6	size	244 non-null	int64

dtypes: category(4), float64(2), int64(1)

memory usage: 7.4 KB

None

Index(['total_bill', 'tip', 'sex', 'smoker', 'day', 'time', 'size'],
dtype='object')

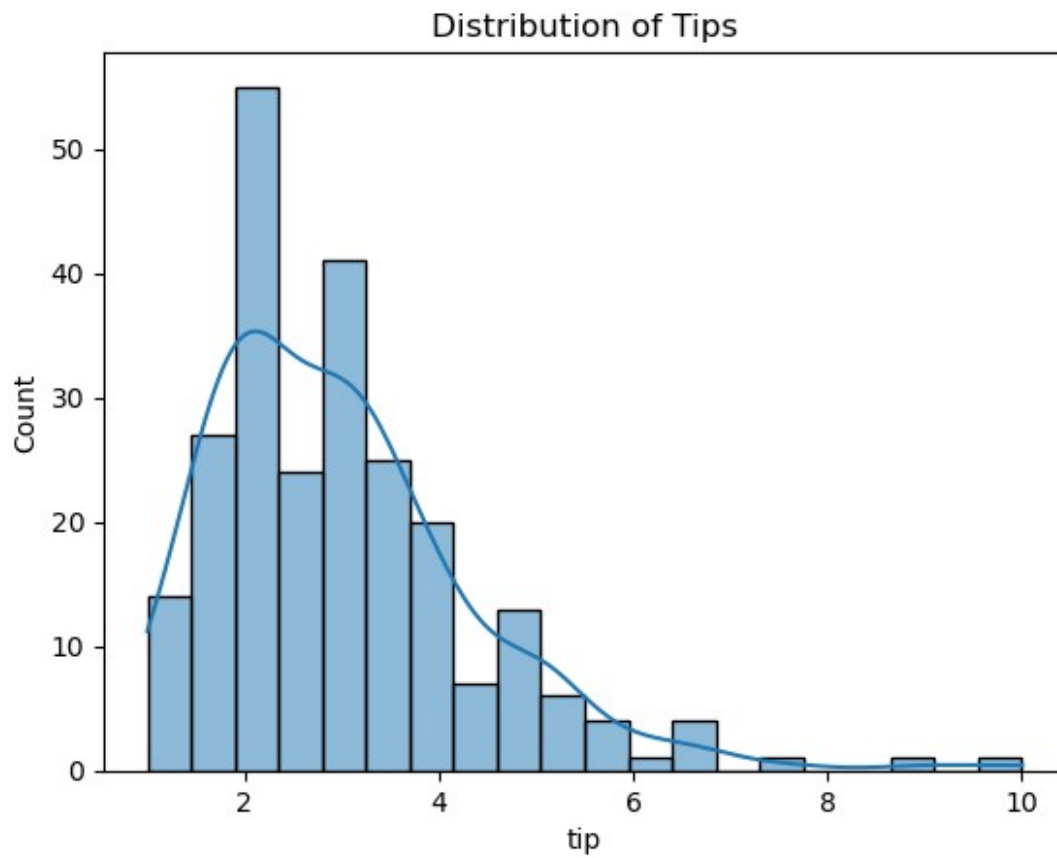
#1.Descriptive Statistics

```
print(df['total_bill'].describe())
```

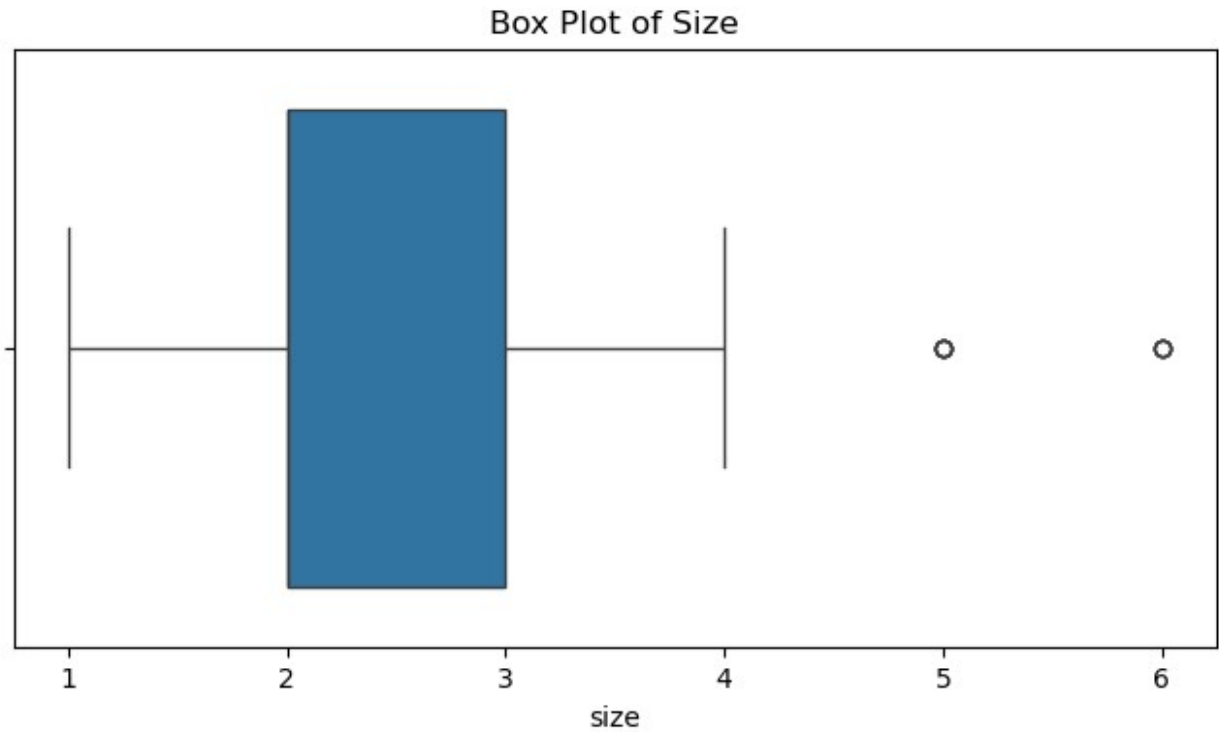
count	244.000000
mean	19.785943
std	8.902412
min	3.070000
25%	13.347500
50%	17.795000
75%	24.127500
max	50.810000

Name: total_bill, dtype: float64

```
sns.histplot(df['tip'], bins=20, kde=True)
plt.title("Distribution of Tips")
plt.show()
```



```
plt.figure(figsize=(8,4))
sns.boxplot(x=df['size'])
plt.title("Box Plot of Size")
plt.show()
```



```
#2.Categorical Variable (value count)
```

```
print(df['day'].value_counts())
```

```
day
```

```
Sat      87
```

```
Sun      76
```

```
Thur     62
```

```
Fri      19
```

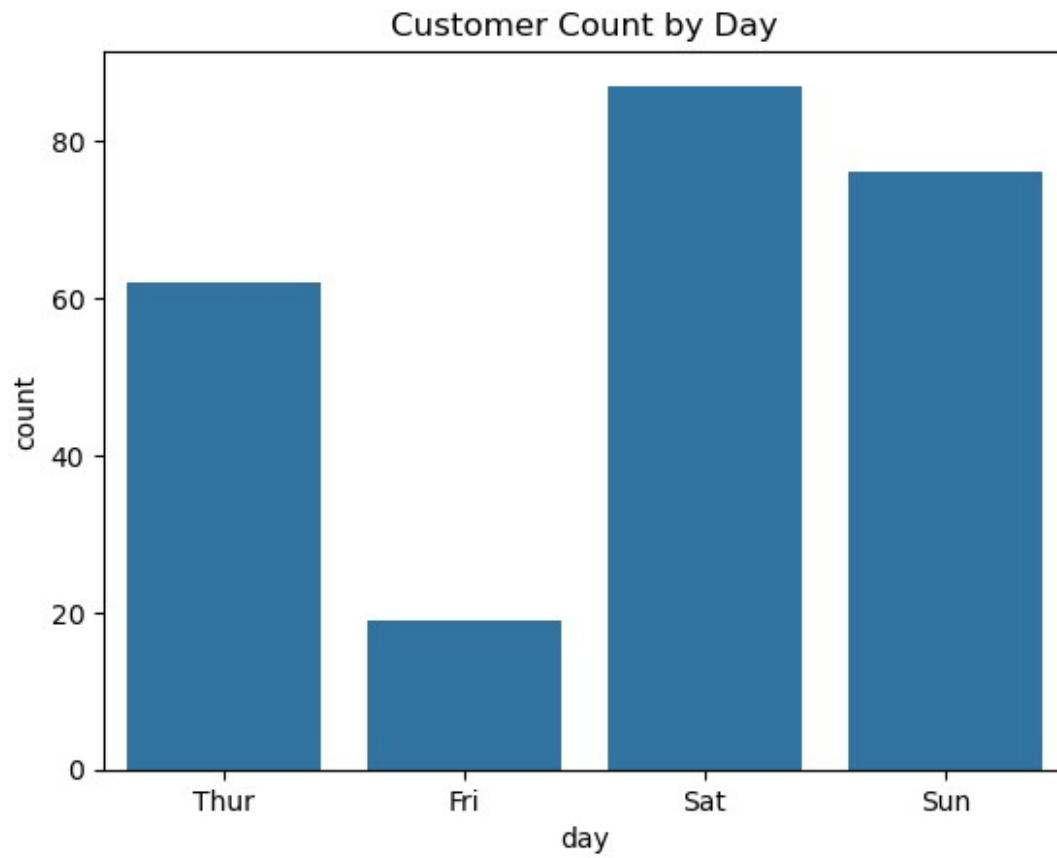
```
Name: count, dtype: int64
```

```
## Count Plot
```

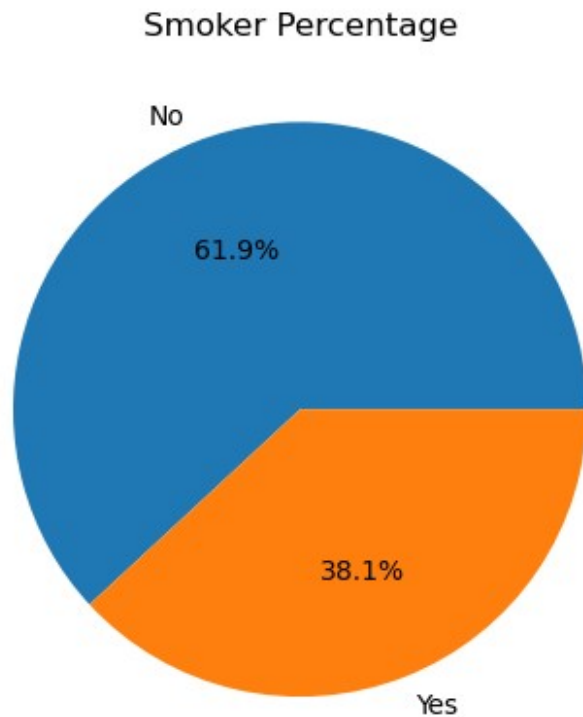
```
sns.countplot(x='day', data=df)
```

```
plt.title("Customer Count by Day")
```

```
plt.show()
```



```
#Pie Chart
df['smoker'].value_counts().plot(
    kind='pie',
    autopct='%1.1f%%'
)
plt.title("Smoker Percentage")
plt.ylabel("")
plt.show()
```



```
#Outliers Detection Using IQR
Q1 = df['total_bill'].quantile(0.25)
Q3 = df['total_bill'].quantile(0.75)

IQR = Q3 - Q1

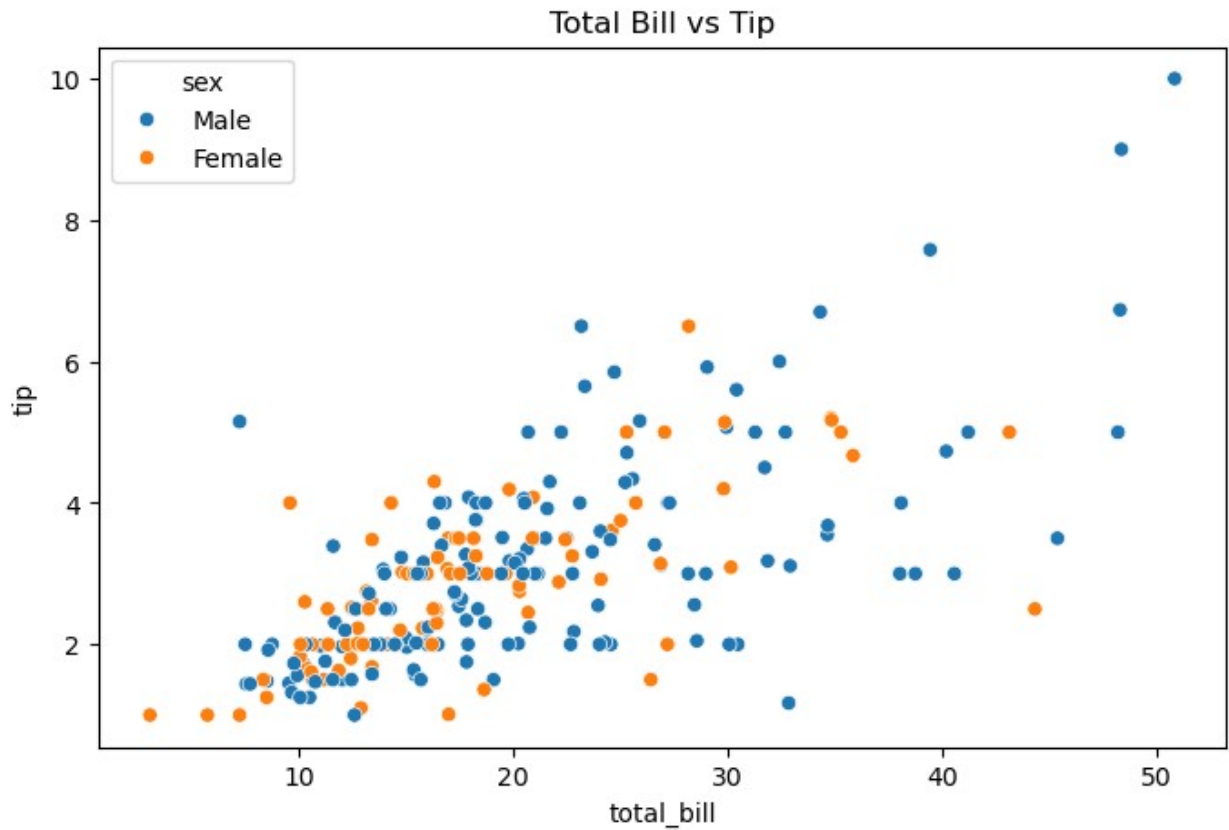
lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

Outliers = df[(df['total_bill'] < lower) | (df['total_bill'] > upper)]

print("Outliers:", len(Outliers))

Outliers: 9

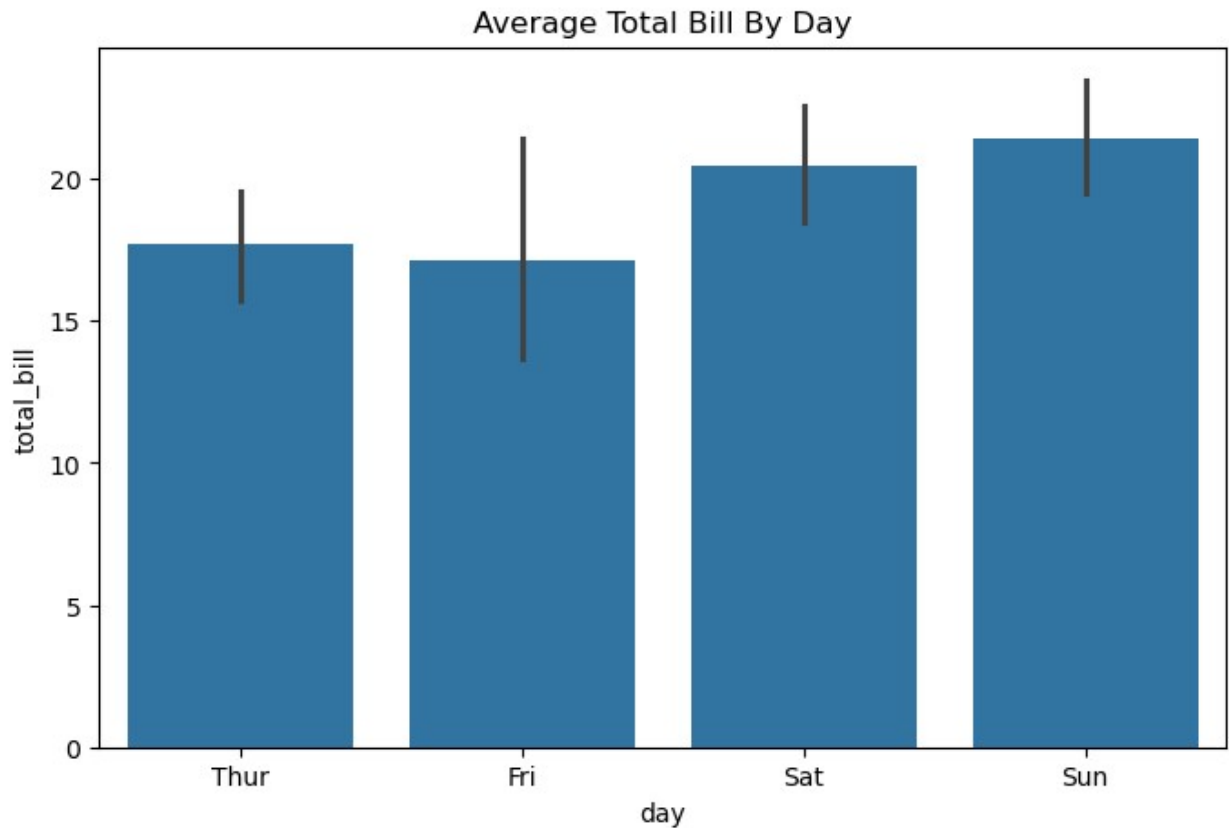
#numeric vs numeric
plt.figure(figsize=(8,5))
sns.scatterplot(
    x='total_bill',
    y='tip',
    hue='sex',
    data=df
)
plt.title("Total Bill vs Tip")
plt.show()
```



```
#Pearson Correlation  
corr = df['total_bill'].corr(df['tip'])  
print(corr)
```

```
0.6757341092113646
```

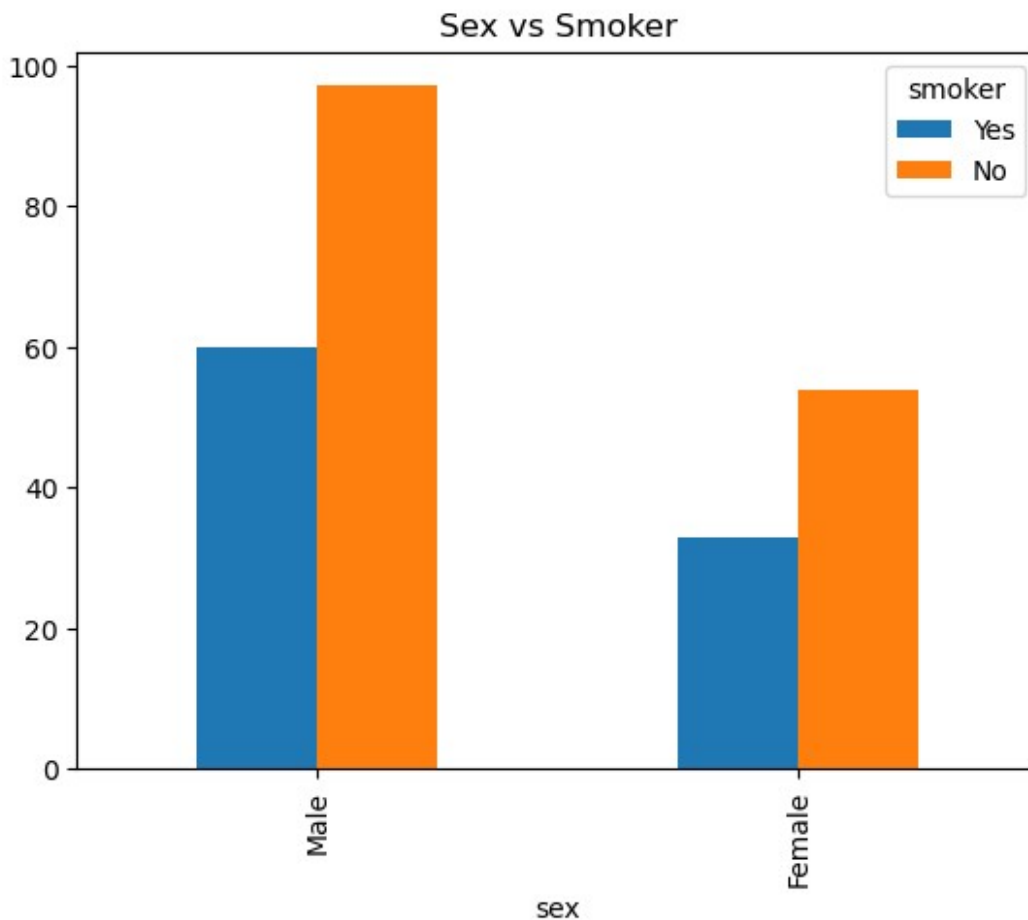
```
# Categorical vs numerical  
#bar chart  
plt.figure(figsize=(8,5))  
sns.barplot(  
    x='day',  
    y='total_bill',  
    data=df  
)  
plt.title("Average Total Bill By Day")  
plt.show()
```



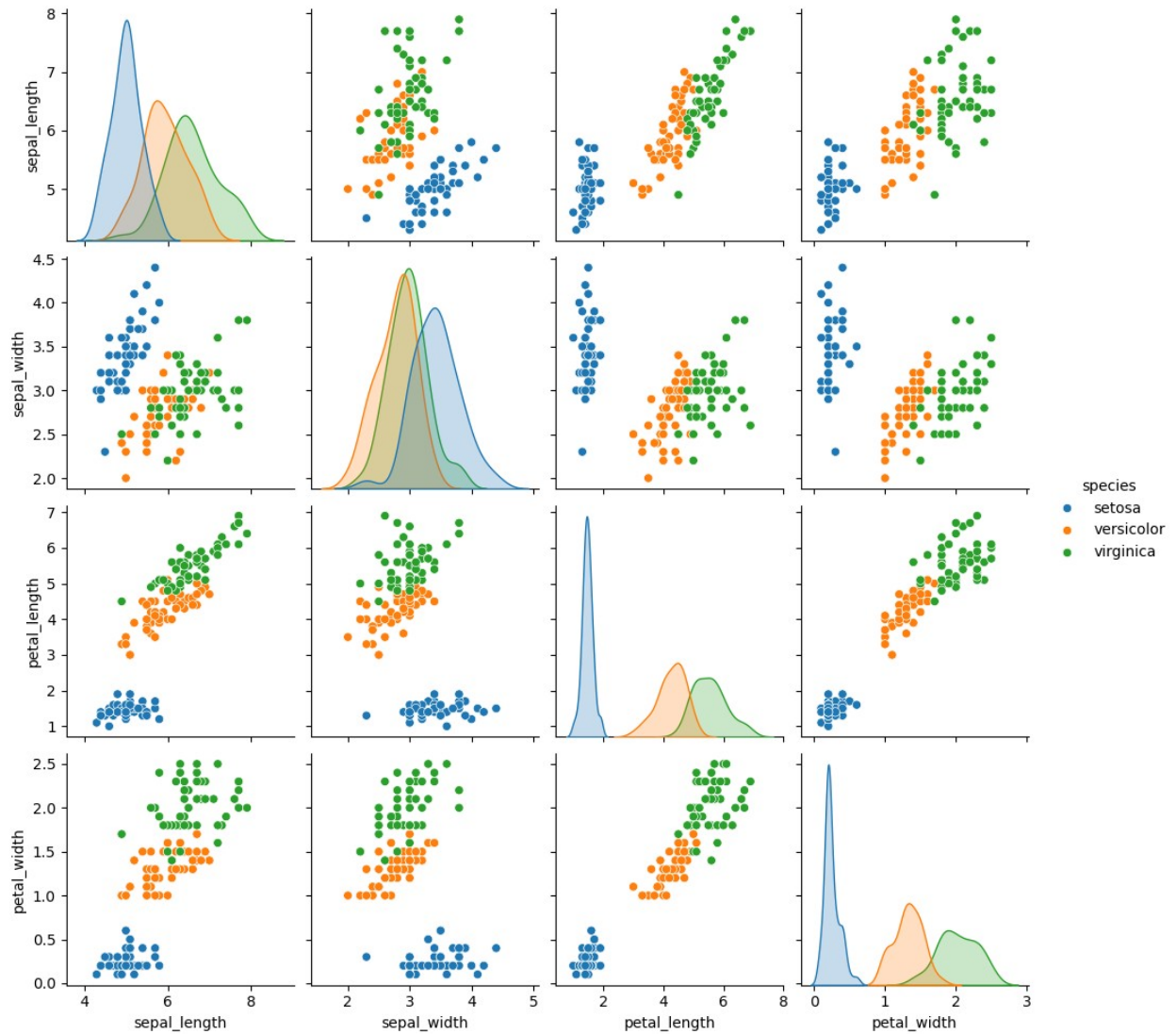
```
#Grouped Statistics
#Mean Bill by Day
print(
    df.groupby('day', observed=False)['total_bill'].mean()
)
#Multiple Statistics
print(
    df.groupby('day', observed=False)["total_bill"]
    .agg(['mean', 'median', 'min', 'max'])
)

day
Thur    17.682742
Fri     17.151579
Sat     20.441379
Sun     21.410000
Name: total_bill, dtype: float64
      mean  median  min  max
day
Thur  17.682742   16.20  7.51  43.11
Fri   17.151579   15.38  5.75  40.17
Sat   20.441379   18.24  3.07  50.81
Sun   21.410000   19.63  7.25  48.17
```

```
#Bar Chart
pd.crosstab(
    df['sex'],
    df['smoker']
).plot(kind='bar')
plt.title("Sex vs Smoker")
plt.show()
```

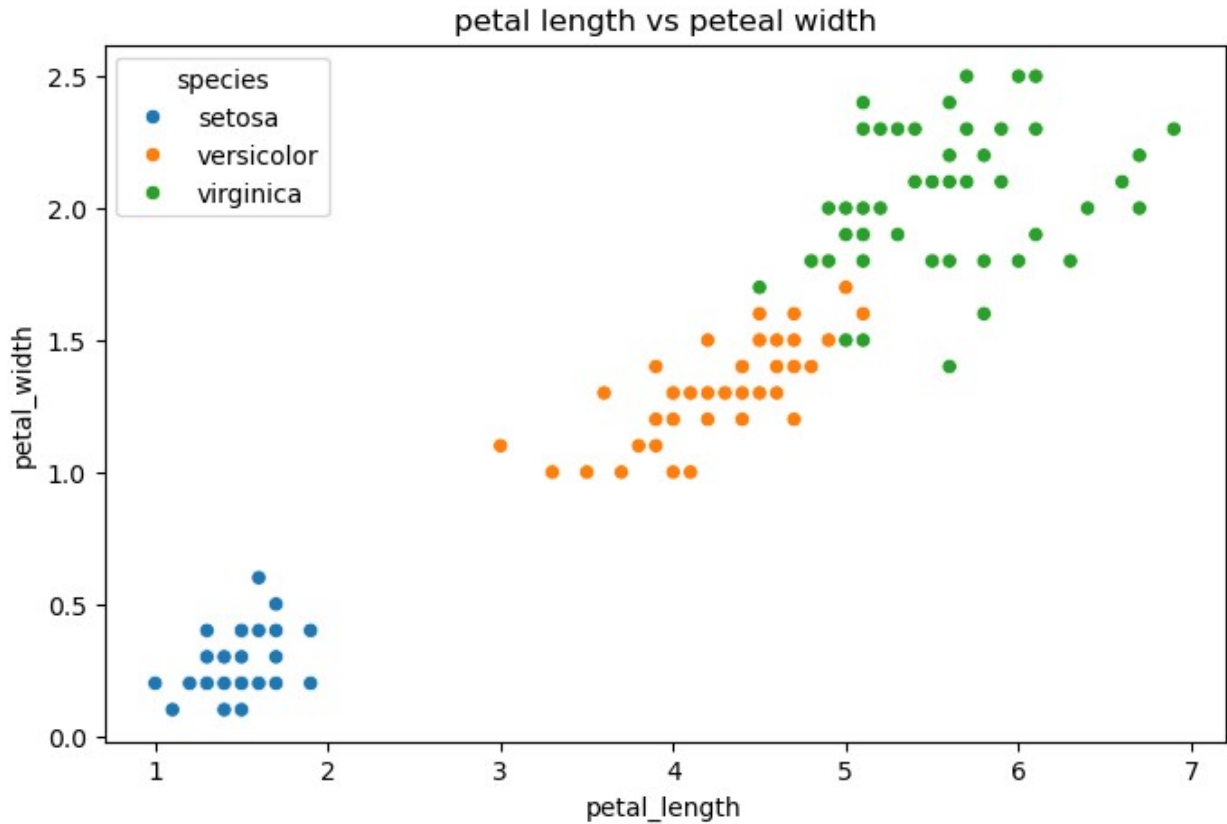


```
#pairplot
import seaborn as sns
import matplotlib.pyplot as plt
df = sns.load_dataset("iris")
sns.pairplot(
    df,
    hue='species'
)
plt.show()
print(df.columns)
```

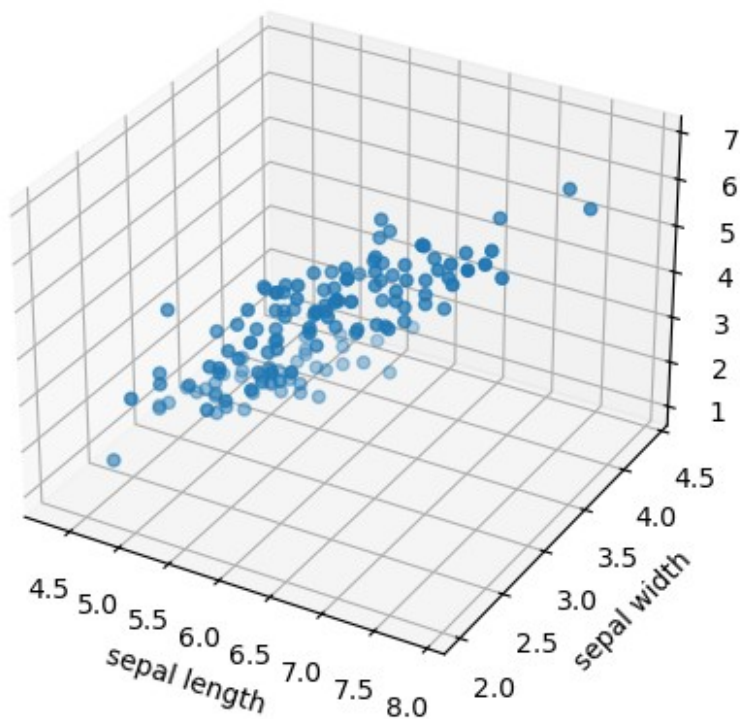


```
Index(['sepal_length', 'sepal_width', 'petal_length', 'petal_width',
      'species'],
      dtype='object')
```

```
#colour -coded scatter plot
plt.figure(figsize=(8,5))
sns.scatterplot(
    x='petal_length',
    y='petal_width',
    hue='species',
    data=df
)
plt.title("petal length vs peteal width")
plt.show()
```



```
#3D Scatter plot
from mpl_toolkits.mplot3d import Axes3D
import matplotlib.pyplot as plt
fig=plt.figure(figsize=(5,5))
ax=fig.add_subplot(111,projection='3d')
ax.scatter(
    df['sepal_length'],
    df['sepal_width'],
    df['petal_length']
)
ax.set_xlabel('sepal length')
ax.set_ylabel('sepal width')
ax.set_zlabel('petal_length')
plt.show()
```



```
import pandas as pd

pd.set_option('future.no_silent_downcasting', True)

df = pd.read_csv("Cricket_data.csv")

df.drop_duplicates(inplace=True)

for col in df.select_dtypes(include="object"):
    df[col] = df[col].fillna(df[col].mode()[0])

df = df.infer_objects(copy=False)

print(df.isnull().sum())
```

season	5
id	0
name	0
short_name	0
description	0
home_team	0
away_team	0
toss_won	0
decision	0
1st_inning_score	0
2nd_inning_score	0

```

winner          0
result          0
start_date      0
end_date        0
venue_id        0
venue_name      0
home_captain    0
away_captain    0
pom            0
points          0
super_over      0
home_overs      13
home_runs       13
home_wickets    13
home_boundaries 13
away_overs      15
away_runs       15
away_wickets    15
away_boundaries 15
highlights      0
home_key_batsman 0
home_key_bowler 0
home_playx1     0
away_playx1     0
away_key_batsman 0
away_key_bowler 0
match_days      0
umpire1         0
umpire2         0
tv_umpire       0
referee         0
reserve_umpire  0
dtype: int64

```

```

import pandas as pd
import seaborn as sns

```

```
df = sns.load_dataset("iris")
```

```

corr_matrix = df.corr(numeric_only=True)
print(corr_matrix)

```

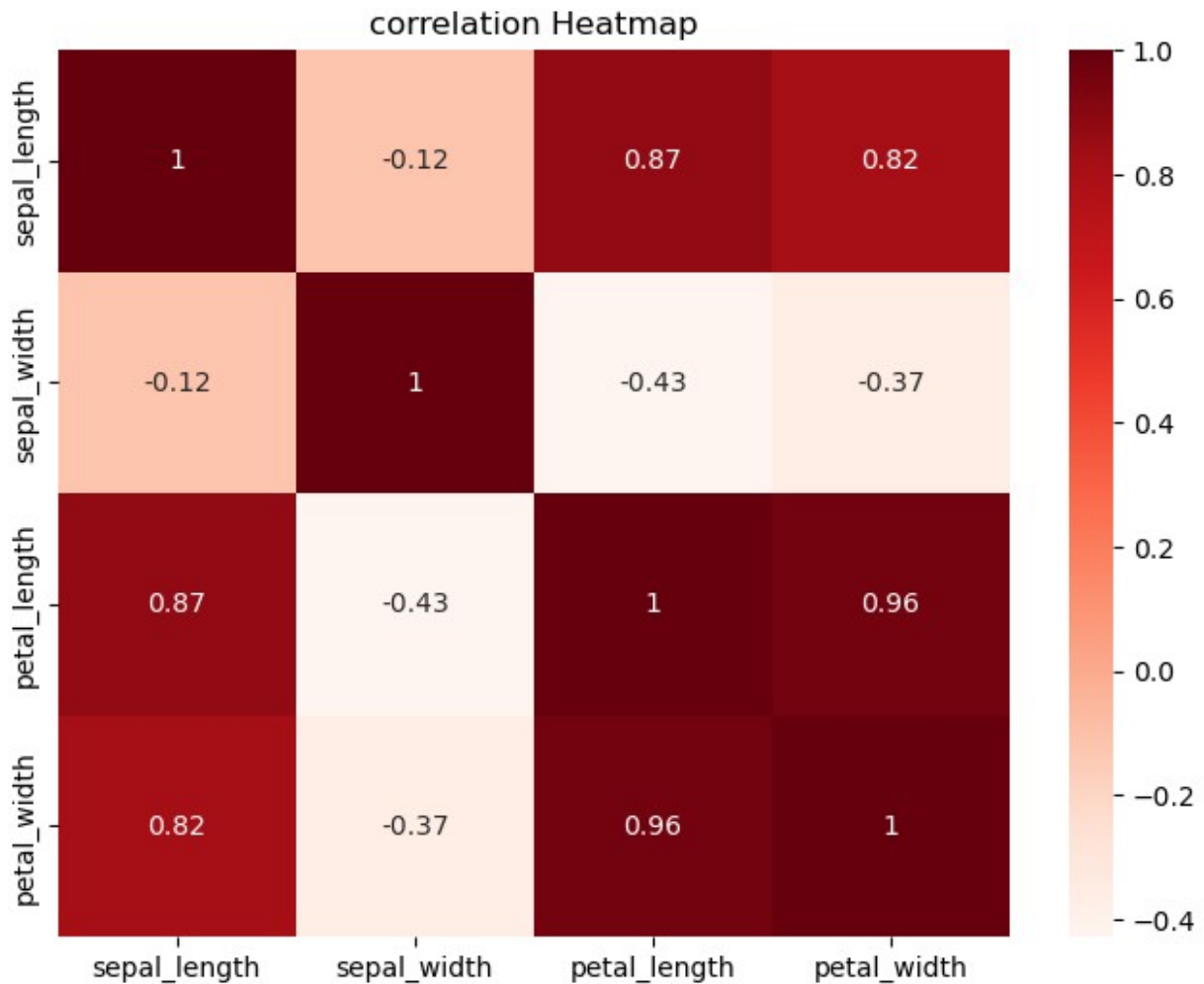
	sepal_length	sepal_width	petal_length	petal_width
sepal_length	1.000000	-0.117570	0.871754	0.817941
sepal_width	-0.117570	1.000000	-0.428440	-0.366126
petal_length	0.871754	-0.428440	1.000000	0.962865
petal_width	0.817941	-0.366126	0.962865	1.000000

```

import pandas as pd
import seaborn as sns

```

```
import matplotlib.pyplot as plt
plt.figure(figsize=(8,6))
sns.heatmap(
    df.corr(numeric_only=True),
    annot=True,
    cmap='Reds'
)
plt.title("correlation Heatmap")
plt.show()
```



```
from statsmodels.stats.outliers_influence import
variance_inflation_factor
x=df.drop("species",axis=1)
vif=pd.DataFrame()
vif["Feature"]= x.columns
vif["VIF"] = [
    variance_inflation_factor(x.values,i)
    for i in range(x.shape[1])
]
```

```

]
print(vif)

      Feature      VIF
0  sepal_length 262.969348
1   sepal_width  96.353292
2  petal_length 172.960962
3   petal_width  55.502060

#Remove a High-VIP Feature
#Suppose we remove petal_length,
X_new = x.drop("sepal_length",axis=1)
vif_after = pd.DataFrame()
vif_after["Feature"] = X_new.columns
vif_after["VIP"] = [
    variance_inflation_factor(X_new.values, i)
    for i in range(X_new.shape[1])
]
print(vif_after)

      Feature      VIP
0   sepal_width  5.856965
1  petal_length 62.071308
2   petal_width 43.292574

#heatmap after removing feature of sepal_length
plt.figure(figsize=(8,6))
sns.heatmap(
    X_new.corr(),
    annot=True,
    cmap='coolwarm'
)
plt.title("correlation Heatmap")
plt.show()

```

